MASTER FINAL SUGGESTION

EEE 337 / EEE 0713 3237: Power System II
3rd Year 2nd Semester Final, SUST

One document that combines everything in the **Power-System-II-Analysis** repository: all 14 commits, all 3 merged pull requests, 4 previous finals, 9 term tests (yours and older batches), 17 lectures, 5 textbook chapters, the syllabus and the course outcomes. Every item has its **source paper** and its **book reference** (book, chapter, article/example/problem number, printed page).

Short form	Meaning
K&N	D.P. Kothari & I.J. Nagrath, Modern Power System Analysis (Ch-2 Inductance, Ch-3 Capacitance, Ch-12 Stability): files Chap_1_and_2.pdf, Chap_12_Nagarath_Stability.pdf
Mehta	V.K. Mehta & Rohit Mehta, Principles of Power System (Ch-8 Mechanical Design of Overhead Lines, Ch-11 Underground Cables): files Chap_8_Mehta.pdf, Chap_11_Mehta.pdf
Stevenson	W.D. Stevenson, Elements of Power System Analysis (seniors marked some questions as Stevenson examples)
Ex / Prob / Art	Worked Example / end-of-chapter Problem (or Tutorial Problem) / Article (section) number. **p.** = printed page number in the book
F21, F22, F24a, F24b	Finals Dec-2021, 2022 (Session 2019-20), 2024 (Session 2020-21), 2024 (Session 2021-22, newest)
TT-1/2/3 (yours)	Your batch's term tests (04_Term_Tests_Questions/)
TT (21-22), TT (20-21)	Older batches' term tests inside 03_Previous_Year_Questions/
Lec-n	Lecture notes in 02_Lecture_Notes/ (teacher remarks such as "For exam", "Derivation very important")

Honest note on "80–90 %": nobody can guarantee the paper. In the 4 past finals, about 70 % of the marks were repeats of older finals or term tests, and almost every remaining mark came from a textbook example or article the teacher listed in class. This suggestion covers **every topic that has been asked at least once** plus **every textbook item the teacher flagged**. If you finish Tier 1 + Tier 2, you should be able to attempt roughly 80–90 % of a typical paper; Tier 1 alone covers roughly 60–65 %.

1. What I checked in the repository (read-only audit)

Nothing in the repository was changed. This PDF is the only new file.

1.1 All commits, first to last

#	Commit	Date	What it did
1	f6880eb	2026-10-24	Initial commit (README)
2	4c173c0	2026-10-24	Added all course material: 5 textbook chapter PDFs, 9 lecture-note PDFs (Lec 1–17), 2 previous-question PDFs, your TT-1/2/3, syllabus, course outcomes
3	64519ef	2026-10-24	Merge of origin/main
4–5	e4a9e5c, fd72356	2026-10-24	README files for the textbook and lecture-note folders
6	7f100d4	2026-10-24	Final_Exam_Suggestion/ folder: ranked suggestion, frequency analysis, solved math bank, derivations guide, short questions, formula sheet
7	ce86c51	2026-10-24	Merge PR #1
8–9	cf6a0c0, 6e737d0	2026-10-24	07_Model_Question_Sets_A_B_C.pdf (3 papers in the new 60-mark pattern, PYQ only) + README order fix
10	8b3a9f5	2026-10-24	Merge PR #2
11	e2d5968	2026-10-24	08_OBE_Model_Question_Sets_1_2_3.pdf (CO and Bloom tagged)
12–13	d1e43ba, dfdbcb2	2026-10-24	09_FINAL_Model_Question_Sets_1_2_3.pdf (20 % PYQ / 60 % textbook + lecture / 20 % external style, every maths item with book reference)
14	83441fc	2026-10-24	Merge PR #3: latest commit on main

1.2 All pull requests (all merged)

PR	Title	Result
mhkamil/Power-System-II-Analysis#1	Add final exam suggestion folder for Power System II	Merged 24 Oct 2026: six Markdown study files + index
mhkamil/Power-System-II-Analysis#2	Add 3 model semester-final question sets (PDF)	Merged: Sets A, B, C from previous-year questions
mhkamil/Power-System-II-Analysis#3	Add OBE and final model question sets	Merged: files 08 and 09 (CO-mapped, with book references)

1.3 Source files analysed

Folder	Files	Used for
01_Textbook_Chapters	K&N Ch-1/2/3 (41 pp), K&N Ch-12 (60 pp), Mehta Ch-8 (43 pp), Mehta Ch-11 (36 pp)	Book example, problem, article and page references (checked against the scans)
02_Lecture_Notes	Lec 1–17 (9 PDFs, about 80 pages)	Teacher's article lists and exam remarks
03_Previous_Year_Questions	F21, F22, F24a, F24b + older TTs with seniors' notes	Repeat counts
04_Term_Tests_Questions	Your TT-1 (PDF), TT-2, TT-3 (photos)	Your batch's questions (very likely to return)
05_Course_Information	Syllabus, CO1–CO5	Scope check: what is in and out
Final_Exam_Suggestion	Files 01–09 from the PRs	Cross-checked; answers re-verified

2. How your paper will most likely look

	Old pattern (F21–F24b)	Your new pattern (60 marks, OBE)
Structure	100 marks, Part A any 2 of Q1–Q3, Part B any 2 of Q4–Q6 (25 marks each)	Part A: Q1, Q2 compulsory + Q3 OR Q4. Part B: Q5, Q8 compulsory + Q6 OR Q7. 10 marks each, 2–4 sub-parts
Part A topics	Ch-2 Inductance, Ch-3 Capacitance, Ch-11 Cables	Same (CO1, CO2)
Part B topics	Ch-8 Overhead lines, Ch-11 Cable faults, Ch-12 Stability, HVDC/FACTS	Same (CO2, CO4, CO5); Q8 is likely stability + HVDC/FACTS
Source mix	About 70 % repeats	About 20 % previous-year, about 60 % textbook examples and lecture notes, about 20 % external-examiner style

What this means: (1) learn every repeated question word for word; (2) for the 60 % textbook share, learn the **worked examples next to the repeated ones** (same formula, different numbers), which are listed in Section 5; (3) for the external 20 %, practise the "justify / compare / why" short answers in Section 6.

2.1 Where the marks come from (weight by chapter)

Chapter	Book	Typical share	Must-know core
Ch-2 Inductance	K&N Ch-2	15–20 %	Transposition, unsymmetrical spacing, composite conductor, bundled (Pheasant, 500 kV), telephone line
Ch-3 Capacitance	K&N Ch-3	10–15 %	Unsymmetrical capacitance derivation, 3.5 m flat spacing, effect of earth
Ch-11 Cables	Mehta Ch-11	20–25 %	Stress derivation, economical size, insulation thickness, 0.645 cm^2 cable, grading, belted cable, Murray, Varley
Ch-8 Overhead lines	Mehta Ch-8	20–25 %	String efficiency (theory + 4 sums), sag (3 types), corona and critical disruptive voltage
Ch-12 Stability	K&N Ch-12	15–20 %	Swing equation, P–δ sudden increase, E′ and P_{max} (healthy / fault / line open), equal-area criterion
HVDC / FACTS / Compensation	Lec 16–17	5–8 %	HVDC definition + economics, types of link, FACTS devices, series/shunt compensation
Power quality	Syllabus only	0 % so far	Definitions only (sag, swell, harmonics, flicker)

3. TIER 1: MUST STUDY (asked 2–5 times; the core of every paper)

Study these first. Every row shows where it was asked and exactly where it is in the book.

3.1 Tier 1 derivations

#	Derivation	Asked	Book reference
D1	**Capacitance of a 3-φ line with unsymmetrical spacing** (transposed): $C_n = 2\pi k / \ln(D_{eq}/r)$, $D_{eq} = \sqrt[3]{(D_{12}D_{23}D_{31})}$	**5×**: F22-2c, F24a-1c, F24b-5d, **your TT-1 Q2**, TT-1 (21-22)	K&N Ch-3, Art 3.6 + **Example 3.4, p.106–107** (full derivation from first principles)
D2	**Inductance of a 3-φ line with unsymmetrical spacing + transposition** ($L = 2\times10^{-7} \ln(D_{eq}/r')$)	F24b-1c, transposition 5×	K&N Ch-2, Art 2.8, p.82 (equilateral, then unsymmetrical + transposition)
D3	**Inductance of composite-conductor lines** (D_m, D_s)	F24a-2c, F24b-4c; Lec-4: "Derivation very important for exam"	K&N Ch-2, Art 2.7; practise with Ex 2.2–2.3, p.81
D4	**Swing equation** ($M\, d^2\delta/dt^2 = P_m - P_e$; per-unit form with H)	F21-6b, F22-6b; Lec-13: "Swing eqn derivation in exam"	K&N Ch-12, Art 12.2 (dynamics of a synchronous machine); Ex 12.1, p.435
D5	**Dielectric stress in a single-core cable**; prove g is maximum at the conductor; $g_{max}/g_{min} = D/d$	**3×**: F21-2b, F22-4d, F24a-3c	Mehta Ch-11, Art 11.9 (dielectric stress); Ex 11.7–11.8
D6	**String efficiency**: potential distribution with equivalent circuit; effect of K = 0.2, K = 0.8, more discs	F24b-3d + theory 4×	Mehta Ch-8, Art 8.6–8.7, p.168
D7	**Most economical conductor size** (d ln(D/d) maximum → $D = e{\cdot}d$, $d = 2V/g_{max}$)	**3×**: F21-3b, F22-3d, F24b-6a	Mehta Ch-11, Art 11.10

3.2 Tier 1 maths (repeated with the same numbers)

#	Problem (short)	Asked	Book reference	Final answer
M1	**Telephone line mutual inductance**: 1-φ line 2.5 m, phone 0.6 m, nearest 20 m, 150 A, 50 Hz	F24a-1d, TT-3 (20-21); new version F24b-1d (3 m, phone 0.8 m, 2 m below, 100 A)	**K&N Ch-2, Prob 2.7, p.93**	M = 0.648 µH/km, V = 30.6 mV/km. F24b version: M = 0.0758 mH/km, V = 2.38 V/km
M2	**Bundled Pheasant** (1,272,000 cmil), d = 45 cm, phases 8 m apart: X per km/mile, per unit (160 km, 100 MVA, 345 kV)	**3×**: F21-1d, F22-2a, F24a-1b	**Stevenson Ex 4.5** (seniors' note); method K&N Art 2.10	0.365 Ω/km, 0.587 Ω/mile, 0.049 pu (60 Hz)
M3	**500 kV, 2-conductor bundle**, 0.5 m, 30 mm dia, 15 m spacing, 50 Hz	F22-1d, TT-1 (21-22)	**K&N Ch-2, Prob 2.9, p.94**	X = 0.346 Ω/km/phase
M4	**3-φ flat spacing 3.5 m**, No. 2/0 Cu (1.05 cm), 110 kV: C_n and charging current	F22-2b, F24b-5b	**K&N Ch-3, Ex 3.2, p.104**	0.00826 µF/km, 0.165–0.17 A/km
M5	**String, 3 discs, top 8 kV, middle 11 kV**: K, line voltage, efficiency	F24b-5c, **your TT-3 Q2**	Mehta Ch-8, Art 8.7 (method of Ex 8.1–8.6, p.171–174)	K = 0.375, V_3 = 18.125 kV, V_L = 64.3 kV, η = 68.3 %
M6	**String, 33 kV, 3 units, m = 11 %**	F24b-3c	Mehta Ch-8 (same method)	5.52, 6.13, 7.41 kV; η = 85.7 %
M7	**String, line unit 17.5 kV, m = 1/8**	F21-4b	**Mehta Ex 8.3** (seniors' note)	V_L = 76.6 kV, η = 84.3 %
M8	**String, 5 discs, 100 kV, m = 0.1**	TT-2 (21-22)	Mehta Ch-8 (same method; Ex 8.6, p.174 is the 4-disc version)	8.00, 8.80, 10.48, 13.21, 17.25 kV; η = 66.9 %
M9	**Sag, unequal towers 30 m / 90 m**, 500 m, T = 1600 kg, 1.5 kg/m	F22-5c	**Mehta Ex 8.23, p.193**	Minimum clearance 23 m; midway 30.7 m
M10	**Sag, towers 40 m / 90 m**, 400 m, T = 2000 kg, 1 kg/m	F21-5c	**Mehta Ex 8.25, p.195**	Midway clearance 55 m
M11	**132 kV support height**: 680 kg/km, 260 m, 3100 kg, SF 2, clearance 10 m	F24a-5c	**Mehta Ex 8.17, p.190**	S = 3.71 m, height 13.71 m
M12	**Sag with wind**: 300 m, 1.3 cm², 1170 kg/km, 4218 kg/cm², SF 5, 122 kg/m²	**Your TT-2 Q1**	**Mehta Ex 8.19, p.191** (200 m, 1.29 cm² in the book)	Slant 20.08 m, vertical 12.0 m
M13	**Murray loop**: 500 m, 1 Ω/1000 m, sound 2.25 Ω/1000 m, ratio 2.75 : 1	**3×**: F22-4c, F24b-1b, TT-3 (21-22)	**Mehta Ex 11.23, p.297**	433.3 m
M14	**Varley loop**: 20 km, 20 Ω/km, S = 200 Ω (your TT: 40 km, 40 Ω/km)	**3×**: F24a-5b, **your TT-2 Q3**, TT-2 (20-21)	**Mehta Ex 11.24, p.297**	15 km (your TT: 37.5 km)
M15	**11 kV cable, 0.645 cm², sheath 2.18 cm, ε_r 3.5**	F24b-2c, TT-2 (21-22)	Mehta Ch-11, Art 11.8–11.9 (method of Ex 11.7)	g_{max} 27.66, g_{min} 11.5 kV/cm; 0.222 µF/km; 0.766 A
M16	**Insulation thickness**: 495 MΩ/km, core 2.5 cm, ρ = 4.5×10¹⁴ Ω-cm	F22-3c, F24b-4b	**Mehta Ex 11.2, p.274**	1.25 cm
M17	**Economical size**: 132 kV 3-φ, g_{max} 60 kV/cm	F24b-6a	Mehta Ch-11, Art 11.10 (Ex 11.9–11.10)	d = 3.59 cm, D = 9.77 cm
M18	**Generator → infinite bus**: x'_d 0.25, j0.1, two lines j0.5: E′ and P_{max} (healthy / 3-φ fault in middle / one line open)	**3×**: F24a-6c, F24b-6d, TT-3 (20-21)	**K&N Ch-12, Ex 12.4, p.444**	E′ = 1.075∠33.9°; P_{max} = 1.79 / 0.693 / 1.264 pu

3.3 Tier 1 theory / short questions

#	Question	Asked	Book reference
S1	**Transposition**: what it is and why (draw the a-b-c rotation over 3 sections)	**5×**: F21-2a, F22-1a, F24a-2b, F24b-1c, your TT-1	K&N Art 2.8, p.82; Fig. 3.13, p.106
S2	**String efficiency**: definition + methods of improving (longer cross-arm, grading, guard ring)	**4×**: F21-4a, F22-5b, F24a-4b, older TT-2	Mehta Art 8.7–8.8
S3	**Critical disruptive voltage**: definition + significance	**3×**: F22-4b, F24a-4a, F24b-1a	Mehta Ch-8, corona: important terms (Art 8.12)
S4	**Corona**: definition, theory of formation, advantages/disadvantages, methods of reducing	**3×**: F21-5a, F22-3a, F24a-2a	Mehta Ch-8, corona articles (Art 8.9–8.14)
S5	**Sudden increase in mechanical input** explained with the P–δ curve (equal-area)	**3×**: F24a-6b, F24b-6c, **your TT-3 Q1**	K&N Ch-12, equal-area criterion articles
S6	**HVDC**: definition, economics (break-even distance), types of link (mono/bi/homopolar)	**3×**: F21-6a, F22-6a, F24b-6b (+ F24b-2a monopolar)	Lec 16–17

#	Question	Asked	Book reference
S7	**Causes of insulation failure** (insulators)	2×: F21-4c, F24a-5a	Mehta Ch-8, insulator articles (Art 8.4–8.5)
S8	**Aluminium vs copper**; composite vs bundled conductor; bundled: what and why	F21-1b/1c, F22-2a, older TT-1	K&N Art 2.10; Mehta Ch-8 conductor materials

4. TIER 2: STUDY NEXT (asked once, or flagged by the teacher)

4.1 Tier 2 maths

Problem	Asked	Book reference	Answer
Double-circuit line a b c a′ b′ c′, 1 m spacing, r = 1.9 cm: L in mH/km/phase	**Your TT-1 Q1**	**K&N Prob 2.12, p.94** (book uses r = 1.5 cm)	0.431 mH/km/phase
Composite conductor: 3 wires of 0.25 cm vs 2 wires of 0.5 cm	Older TT-1	Stevenson Ex 4.2; K&N Ex 2.2–2.3, p.81	$L = 14.7\times10^{-7}$ H/m
Partridge 1-φ, 20 ft: capacitive susceptance per mile	Older TT-1	Stevenson (Partridge, 60 Hz)	2.55 μS/mile
Effect of earth: two conductors 0.229 in, 10 ft apart, 25 ft above ground	F21-2c	K&N Art 3.7; Ex 3.1, p.103	Use $r(1 + D^2/4h^2)^{½}$ correction
33 kV cable, d = 1 cm, D = 4 cm: g_{max}, g_{min}	F24a-2d	**Mehta Ex 11.7**	47.6 / 11.9 kV/cm
g_{max} 40, g_{min} 10 kV/cm, d = 2 cm: thickness, voltage	F24a-4c	**Mehta Ex 11.8**	3 cm, 55.45 kV
Graded cable, 66 kV, ε 5 and 3, 1 cm each, core 2 cm	F24b-4d	Mehta Art 11.12, p.281 (capacitance grading)	48.2 and 40.2 kV/cm
3-core belted cable 12.6 μF / 7.4 μF, 66 kV	F24a-3d	**Mehta Ex 11.18, p.290**	C_N = 9 μF, I_c = 107.7 A
33 kV, 4 km, three single-core cables (2.5 cm, 0.5 cm, $ε_r$ 3)	F21-3c	**Mehta Ex 11.6**	1.98 μF, 11.87 A, 678 kVAR
66 kV, 1 km, core 10 cm, insulation 7 cm, $ε_r$ 4	F21-2d	**Mehta Ex 11.5, p.276**	0.254 μF, 3.04 A
11 kV 1-φ cable, 2.5 km, d 20 mm, sheath r 15 mm, $ε_r$ 2.4	Older TT-2	Mehta Ch-11 (capacitance examples)	0.823 μF, 2.84 A, 31.3 kVAR
Insulation resistance, 3 km, r 12.5 mm, t 10 mm, ρ 5×10^{12} Ω-m	F21-3a	Mehta Ch-11, Tutorial 11.3 (seniors' note)	155.9 MΩ
Permissible current loading (core 30 mm, dielectric 40 mm, 55 °C)	**Your TT-2 Q2**	**Mehta Ex 11.20, p.292**	≈ 580 A
Corona: 132 kV, 1.956 cm, corona above 210 kV, g_0 30 kV/cm: spacing	F21-5b	**Mehta Ex 8.14**	≈ 3.38 m
Power-angle equation: x'_d 0.2, j0.1, lines j0.4 ‖ j0.4	F22-6c	K&N Ch-12 (Ex 12.4 type)	$P_e = 2.10 \sin δ$

4.2 Tier 2 derivations and theory

Topic	Asked / flagged	Book reference
Inductance of a 1-φ two-wire line, $L = 4\times10^{-7} \ln(D/r')$	F21-1a	K&N Art 2.4
Flux linkages of one conductor in a group	F22-1c	K&N Art 2.6
3-φ equilateral spacing: prove $L_a = L_b = L_c$; capacitance with equilateral spacing	F22-1b	K&N Art 2.8, p.82; Art 3.5, p.99
Capacitance of a two-wire line ($C_n = 2C_{ab}$)	Syllabus	K&N Art 3.4, p.98
Effect of earth on capacitance (method of images)	F24b-5a	K&N Art 3.7
Grading of cables: why unequal stress is bad; capacitance grading with 3 dielectrics; intersheath grading	F24b-2d	Mehta Art 11.11–11.13, p.281
Capacitance of a 3-core cable, measurement of C_e and C_c	F24a-3d (maths)	Mehta Art 11.14–11.15, p.288
Construction of a 3-core cable with figure	F24b-2b; Lec: "11.1 figure important"	Mehta Art 11.1, p.268
Types of cable faults; Murray and Varley loop derivations	Loop sums 3× each	Mehta Art 11.20–11.23, p.293–295
Thermal resistance, permissible current loading	F22-4a, your TT-2	Mehta Art 11.16–11.19, p.291
Sag: equal supports ($wl^2/8T$), unequal supports, wind and ice loading; stringing chart	F22-4a, F24b-4a	Mehta Art 8.15–8.16, p.187–189
Types of insulators; pin-type breakdown; line supports for ≤ 11 kV; main components of an overhead line	F24b-3a/3b, F22-5a	Mehta Art 8.1–8.5, p.164

Topic	Asked / flagged	Book reference
Equal-area criterion (derivation, applications); power angle	F24a-6a	K&N Ch-12, transient stability articles
Multi-machine swing equation (H_{eq}), factors affecting stability	Syllabus	K&N Ch-12
FACTS devices and applications; series and shunt compensation	F24a-3b	Lec 16–17
Skin effect and proximity effect; define inductance	F24a-3a, F24a-1a	K&N Art 2.2, 2.12

4.3 Tier 3 (only if time is left) and what to skip

Classification and laying of cables · line supports and conductor materials · steady-state stability (K&N Ex 12.2, p.440; Ex 12.6, p.447) · numerical solution of the swing equation · power-quality definitions. **Skip:** travelling-wave pulse on a coaxial line (F21-6c, outside the current syllabus) and K&N Example 3.3 (teacher: "Not necessary").

5. The 60 % textbook share: examples to practise next to the repeats

The new pattern takes about 60 % from textbook examples, problems and lecture notes. These are the textbook items closest to the repeated questions (same formula, new numbers). All were also used in the model papers (files 07–09), so you can practise them there.

Chapter	Practise these (book, number, page)
K&N Ch-2	Ex 2.2 and 2.3, p.81 (composite conductor) · Prob 2.7, p.93 (telephone line) · Prob 2.8, p.93 · **Prob 2.9, p.94** (500 kV bundle) · Prob 2.10, p.94 · **Prob 2.12, p.94** (double circuit) · MCQ 2.9–2.12, p.95 (bundles, transposition)
K&N Ch-3	Ex 3.1, p.103 (effect of earth) · **Ex 3.2, p.104** (3.5 m flat spacing) · **Ex 3.4, p.106** (unsymmetrical derivation) · Prob 3.3, p.110 · Prob 3.4, p.110 · Prob 3.8, p.111
Mehta Ch-11	Ex 11.1, p.273 · **Ex 11.2, p.274** · **Ex 11.5, p.276** · Ex 11.6 · Ex 11.7 · Ex 11.8 · Ex 11.9, p.279 · Ex 11.12, p.282 · Ex 11.13, p.283 · Ex 11.15, p.285 · Ex 11.17, p.289 · **Ex 11.18, p.290** · Ex 11.19, p.290 + Tutorial 1–3, p.290 · **Ex 11.20, p.292** · Ex 11.21, p.296 · Ex 11.22, p.297 · **Ex 11.23–11.24, p.297** · Tutorial 3 after 11.24, p.298
Mehta Ch-8	Ex 8.1, p.171 · Ex 8.3 · Ex 8.5, p.173 · Ex 8.6, p.174 · Ex 8.8, p.176 · Ex 8.9, p.177 · Ex 8.11–8.12, p.179 · Ex 8.13, p.184 · Ex 8.14 · Ex 8.15–8.16, p.185 · **Ex 8.17, p.190** · **Ex 8.19, p.191** · Ex 8.22, p.192 · **Ex 8.23, p.193** · Ex 8.24, p.194 · **Ex 8.25, p.195**
K&N Ch-12	Ex 12.1, p.435 (H, stored energy, acceleration) · Ex 12.2, p.440 (steady-state limit) · **Ex 12.4, p.444** (E′ and P_{max}) · Ex 12.6, p.447

Bold = the example is the exact source of a past exam or TT question.

6. The 20 % external-examiner share: likely "justify / compare" questions

Likely question	Key points to write
Overhead line or underground cable for 11 kV in a city? Justify.	Cable: safety, appearance, no lightning or storm faults, low maintenance; but 5–10× cost, hard fault location, charging current
Why are bundled conductors used in EHV lines?	Lower reactance (larger GMR), higher capacitance, less corona loss and radio interference, higher surge impedance loading
Why is transposition needed? What if the line is not transposed?	Unequal L and C per phase → unbalanced voltage drops, interference in nearby telephone lines
Guard ring or one more disc to improve a poor string?	Guard ring equalises voltage (reduces the top-to-bottom gradient); an extra disc raises total withstand but lowers efficiency
Why must a cable be graded? Why is intersheath grading rarely used?	Uniform stress → smaller cable for the same voltage; intersheath: damage risk, charging current in intersheaths, complex to set potentials
Effect of fault clearing time on stability	Equal-area criterion: faster clearing → smaller accelerating area → stable; critical clearing angle
When is HVDC better than HVAC?	Long distance beyond the break-even distance (≈ 500–800 km overhead, ≈ 50 km cable), asynchronous links, no charging current, no stability limit
How do series capacitors increase power transfer?	$X' = X - X_C \rightarrow P_{max} = V_s V_r / X'$ rises; better voltage regulation and stability; risk of sub-synchronous resonance
Hazards of a line passing near a telephone line	Induced voltage $V = \omega M I$ (M1 type), noise; remedies: transposition of both lines, more distance, screening

7. One-page formula sheet

Topic	Formula
Inductance	$r' = 0.7788\ r \cdot L = 2\times10^{-7}\ \ln(D_m/D_s)$ H/m per conductor · 1-φ loop $L = 4\times10^{-7}\ \ln(D/r') \cdot D_{eq} = \sqrt[3]{(D_{ab}D_{bc}D_{ca})}$
Bundle GMR	2 conductors: $\sqrt{(D_s d)}$ · 3: $\sqrt[3]{(D_s d^2)}$ · 4: 1.09 $\sqrt[4]{(D_s d^3)}$ (use r instead of D_s for capacitance)
Telephone line	$M = 2\times10^{-7}\ \ln[(D_{14}D_{23})/(D_{13}D_{24})]$ H/m · $V = \omega MI$
Capacitance	$C_n = 2\pi\varepsilon_0 / \ln(D_{eq}/r)$ F/m = 0.0242/log(D_{eq}/r) μF/km · $C_{ab} = \pi\varepsilon_0/\ln(D/r) \cdot I_c = \omega C_n V_{ph}$ · earth effect: $r \to r(1 + D^2/4h^2)^{½}$
Cable	$C = 2\pi\varepsilon_0\varepsilon_r/\ln(D/d) = \varepsilon_r/(41.4\ \log(D/d))$ μF/km · $g_{max} = 2V/(d\ \ln(D/d)) \cdot g_{min} = 2V/(D\ \ln(D/d)) \cdot R_{ins} = (\rho/2\pi l)$ ln(D/d) · economical: D = 2.718d, $d = 2V/g_{max}$ (V peak, phase)
Grading	$\varepsilon_1 d\ g_1 = \varepsilon_2 d_1 g_2 \cdot V = (g_1 d/2)\ln(d_1/d) + (g_2 d_1/2)\ln(D/d_1)$
3-core cable	C_1 (all cores to sheath) = $3C_e \cdot C_2$ (one core to others + sheath) = $C_e + 2C_c \cdot C_N = C_e + 3C_c$
Thermal	$S_{dielectric} = (k/2\pi)\ \ln(r_2/r_1) \cdot I = \sqrt{(t / (nRS))}$
Loop tests	Murray: X = [Q/(P+Q)] (R + X) · Varley (P = Q): X = (loop − S)/2 · distance = X / (Ω per length)
String	$V_2 = V_1(1 + K) \cdot V_3 = V_1(1 + 3K + K^2)$ · general $V_{n+1} = V_n + K(V_1 + \ldots + V_n) \cdot \eta = V_{ph}/(n \times V_{bottom})$
Sag	$S = wl^2/(8T)$, T = breaking load/SF · unequal: $x_1 = l/2 - Th/(wl)$, $S = wx^2/(2T)$ · wind: $w_t = \sqrt{((w + w_{ice})^2 + w_w^{\ 2})}$, vertical sag = $S \cos\theta = S{\cdot}w/w_t$
Corona	$V_c = m_0 g_0 \delta\ r\ \ln(d/r)$ kV/phase, g_0 = 21.1 kV/cm rms (30 peak) · δ = 3.92b/(273 + t)
Stability	$M\ d^2\delta/dt^2 = P_m - P_e \cdot (2H/\omega_s)\ d^2\delta/dt^2 = P_m - P_e$ (pu) · $P_e = (E'V/X)\ \sin\delta \cdot E' = V_t + jx'_d I$ · equal area: $\int(P_m - P_e)d\delta = 0$

8. Study plan

Day	Morning	Afternoon	Night (revise)
Day 1: Ch-2 + Ch-3	D1, D2, D3 + transposition (K&N Ex 3.4, p.106)	M1 telephone, M2 Pheasant, M3 500 kV, M4 3.5 m, TT-1 double circuit	Short: S1, S8, skin effect, bundled
Day 2: Ch-11 Cables	D5 stress, D7 economical size, grading derivation	M13 Murray, M14 Varley, M15, M16, M17, belted, permissible current	Cable construction figure, faults, thermal resistance
Day 3: Ch-8 Overhead	D6 string theory + S2	M5–M8 strings (one formula), M9–M12 sag, corona spacing	S3 CDV, S4 corona, insulator types
Day 4: Ch-12 + HVDC	D4 swing equation, equal area, S5	M18 (E′, three P_{max}, draw three curves), power-angle eqn	S6 HVDC, FACTS, compensation
Day 5: Mock	Write one paper from 09_FINAL_Model_Question_Sets in 2 hours	Check answers against 03_Solved_Math_Bank.md	Formula sheet (Section 7) twice

Only one day left? D1 + S1 → M1, M2 → D5 + M13 + M14 → S2 + M5 + M9 → D4 + S5 + M18 → S6 → formula sheet. That set alone touches every question of every past final.

Exam-hall tips: in Part A choose the question with the unsymmetrical derivation and the telephone/bundled maths; in Part B choose string/sag and stability. Write given → formula → substitution → answer with units. For E'/P_{max}, always draw the reactance diagram for each condition. For string sums, always draw the equivalent circuit (it is often worth marks on its own).

9. Where to find full solutions in the repository

Need	File
Step-by-step solutions to every Tier 1/Tier 2 maths item	Final_Exam_Suggestion/03_Solved_Math_Bank.md
Derivation skeletons (D1–D7 and Tier 2)	Final_Exam_Suggestion/04_Derivations_Guide.md
Ready short answers	Final_Exam_Suggestion/05_Short_Questions_and_Definitions.md
Proof of the repeat counts	Final_Exam_Suggestion/02_Question_Frequency_Analysis.md
Practice papers in the new 60-mark pattern	07, 08 and 09 (use 09 first)